

Vinoy Kumar

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EDUCATION

Arizona State University

Bachelor of Science in Computer Science

Tempe, AZ

May 2026

Relevant Coursework: Operating Systems, Computer Networks, Computer Systems Security

EXPERIENCE

Software Engineer Intern

August 2025 - December 2025

Qualaces

Ann Arbor, MI

- Developed a HIPAA-compliant multimodal **vision-language** pipeline using **Google ML Kit** and **Ollama** to parse medication bottles and prescriptions, deployed across over 50 clinics.
- Built a RESTful API and dashboard with **Supabase** and **Amazon Web Services' ECS (Fargate)** to manage PHI, supporting secure CRUD operations for therapists and clinics.

Systems Research Assistant

May 2025 - September 2025

SCAI, Arizona State University

Tempe, AZ

- Developed benchmarking framework in **QEMU** to profile **VirtIO** device performance to identify latency bottlenecks in guest-to-host disk I/O path.
- Integrated **LVM** based virtual disks in QEMU + KVM to improve I/O throughput while maintaining data integrity in virtualized environments.

Software Engineer Intern

September 2022 - August 2023

DevilSec

Tempe, AZ

- Revamped an **SPA** using **React**, **Express** and **JavaScript**, porting it to mobile devices and increasing Monthly Active Users (MAU) by 22%.
- Implemented a **PostgreSQL** database using **AWS RDS** with hashed credential storage to manage the organization's member list and login functionality, increasing user personalization and secure data management.

Linux Kernel Developer (Open Source)

May 2025 - September 2025

Linux Foundation

- Contributed bugfix patches to Linux kernel staging tree subsystems such as **USBip**, **PCIe**, and **VirtIO-blk**, enhancing driver reliability and code quality by adhering to upstream conventions and utilizing **checkpatch.pl**.

PROJECTS

CodeSync - Distributed Code Execution Service | *React, Express, Websockets, Docker, Redis, AWS, Gen AI*

- Built real-time collaborative coding platform able to support 100+ concurrent users with **WebSocket**-based **microservices** architecture.
- Achieved sub-2s execution latency for 500+ sandboxed **Docker** code runs using **Redis** job queue and deployed on **AWS** with **CI/CD**
- Integrated **Claude AI** for live pair programming, enabling collaborative **code generation** and debugging in shared sessions

NES Emulator | *C++, Graphics Rendering, Hardware Emulation*

github.com/fullmoonemptysun/witNES

- Implemented cycle-accurate 6502 CPU emulator in **C++** with full 256-instruction set and precise clock timing.
- Developed a **Picture Processing Unit** graphics renderer synchronized with CPU cycles, producing 95% accurate NES frame output.

Cross - Platform C Compiler | *C++, Compilers, LLVM, x86 Assembly*

github.com/fullmoonemptysun/ccompiler

- Built complete **C compiler** from scratch in **C++** supporting pointers, structs, arrays, and function calls with full type checking and **semantic analysis**.
- Implemented multi-pass **optimization** pipeline with constant folding, dead code elimination, and loop unrolling.
- Integrated **LLVM** backend for cross-platform code generation with register allocation and native machine code output.

SKILLS

Languages: C, C++, Python, Bash, JavaScript, SQL, C#, Java, Kotlin

Frameworks: React, Node.js, Express, .NET, Jest

Backend & Infrastructure: Redis, Kubernetes, Docker, Jenkins, PostgreSQL, AWS, gRPC

Developer Tools: Git, GitHub actions, Linux/Unix

Concepts: Full Stack Development, Distributed Systems, Networking (TCP/IP, OSI model), Generative AI, CI/CD